

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

CS1 - Actuarial Statistics (Paper A)

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** Which of the following best describes the purpose of descriptive analysis? [1]
- A) To make specific conclusions about the data
 - B) To summarize and present data in a simpler, more understandable format
 - C) To test hypotheses using large data samples
 - D) To predict future events based on historical data
 - E) To eliminate outliers and anomalies in the data
- Q. 2)** What is the primary difference between measures of central tendency and measures of dispersion in descriptive analysis? [1]
- A) Central tendency describes the "spread" of values, and dispersion describes the "average" value
 - B) Central tendency describes the "average" value, and dispersion describes the "spread" of values
 - C) Both central tendency and dispersion describe the "spread" of values
 - D) Central tendency refers to the maximum value, while dispersion refers to the minimum value
 - E) Central tendency is related to outliers, and dispersion is related to normal distribution
- Q. 3)** X has a Poisson distribution with mean 11 and Y has a Poisson distribution with mean 19. If $\text{COV}(X, Y) = -24$, calculate the variance of Z where $Z = 2X - 3Y + 4$. [2]
- A) 215
 - B) 503
 - C) -73
 - D) 359
 - E) 367
- Q. 4)** A standard six-sided die is rolled 150 times, with each roll resulting in one of the outcomes: 1, 2, 3, 4, 5, or 6. Let S represent the total sum of the outcomes from these 150 rolls. Calculate the approximate probability that S lies between 450 and 500, inclusive. [3]
- A) 0.00
 - B) 0.50
 - C) 0.1208
 - D) 0.1149
 - E) 0.2451
- Q. 5)** A random sample of size n is taken from a gamma distribution with parameters $\alpha = 9$ and $\lambda = \frac{3}{\theta}$. The sample mean is $\bar{X}$, and θ is to be estimated.
- i)** Determine the method of moments estimator (MME) of θ . [2]
- A) $\frac{1}{3\bar{X}}$
 - B) $\frac{27}{\bar{X}}$
 - C) $\frac{\bar{X}}{27}$
 - D) $\frac{\bar{X}}{9}$
 - E) $\frac{\bar{X}}{3}$

- ii) Find the bias of the MME determined in part (i). [2]
- (A) $\frac{3}{n}$
- (B) 2θ
- (C) 0
- (D) $\frac{\theta}{n}$
- (E) $\frac{\theta}{3}$
- iii) Determine the mean square error of the MME of theta. [2]
- (A) $\frac{3\theta^2}{n}$
- (B) $\frac{\theta^2}{9n}$
- (C) $\frac{9\theta^2}{n}$
- (D) $\frac{1}{9n\theta^2}$
- (E) $\frac{\theta^2}{n}$

Q. 6) The random variables X and Y are related as follows:

$$X|Y=y \sim N(y + 3, 0.01y^2 + 5).$$

Y has a N (100, 60) distribution.

Derive the unconditional variance of X, V(X)

[3]

- A) 165
- B) 168.6
- C) 66
- D) 165.60
- E) 105.6

Q. 7) XYZ is a small insurance company that specialises in cyber insurance, protecting insureds from losses due to cybercrime. The time T, measured in days, until the next claim is reported under a portfolio, follows an exponential distribution with a mean of 5 days.

- i) What is the probability that no claim is reported in the next two days? [2]
- A) 0.05
- B) 0.3297
- C) 0.7297
- D) 0.8187
- E) 0.6703
- ii) The median of a random variable is defined as the value at which the probability of the variable being less than or equal to it is 0.5. Find the median time until the next claim arises. [2]
- A) 5 days

- B) 3.466 days
- C) 1.505 days
- D) 2.5 days
- E) 0.1386 days

- iii) Suppose XYZ tracks 15 similar portfolios. Now let $T_1, T_2, \dots, T_{15}$ be the times (in days) until the next claim for each portfolio, where each T_i , $i = 1, \dots, 15$, follows an exponential distribution with mean 5, independently of all others.

Calculate, approximately, the probability that the total of all 15 times which elapse until a claim arises on each of the portfolios exceeds 60 days.

[3]

- A) 0.772
- B) 0.781
- C) 0.0312
- D) 0.7257
- E) 0.6997

- Q. 8)** A random variable X is believed to have probability density function, $f(x)$, where:

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

To test the null hypothesis $\lambda = 1$ against the alternative hypothesis $\lambda = 2$, a single value is observed. If this value is less than 0.1, H_0 is rejected.

- i) Calculate the size of the test.

[2]

- A) 0.0952
- B) 0.1813
- C) 0.9048
- D) 0.01
- E) 0.99

- ii) Calculate the power of the test.

[2]

- A) 0.0952
- B) 0.1353
- C) 0.1813
- D) 0.8187
- E) 1.00

- Q. 9)** An insurer has collected data about the systolic blood pressure (in mmHg) of 250 females between the ages of 18 and 40. The results are shown in the following table.

Systolic blood pressure (in mmHg)	<100	100-200	120-140	>140
Observed frequency	12	124	78	36
Relative frequency	6p	60p	36p	23p

A statistician computed the estimate of p using the maximum likelihood estimation method and the value of $\hat{p}$ resulted to be 0.008.

You are required to perform a goodness-of-fit test at 5% level of significance to assess whether the proposed distribution is a good fit for the observed data.

- i) Calculate the test statistic to decide whether the suggested distribution is appropriate for the observed data. [4]
- A) 6.302
B) 2.391
C) 0.633
D) 2.807
E) 2.791
- ii) What is the critical value for the given test? Do you accept or reject the hypothesis? [2]
- A) 7.815, Reject the null hypothesis that the model is a good fit.
B) 6.251, Reject the null hypothesis that the model is a good fit.
C) 7.815, Accept the null hypothesis that the model is a good fit.
D) 6.251, Accept the null hypothesis that the model is a good fit.
E) 11.345, Accept the null hypothesis that the model is a good fit.

Q. 10) Let X be a random variable with probability density function:

$$f(x) = \frac{4}{5}e^{-x} + \frac{2}{5}e^{-2x} \quad x \geq 0$$

- i) Find the moment generating function of the above function. [3]
- A) $Mx(t) = \frac{1}{(1-t)}$ for $t < 1$
B) $Mx(t) = \frac{1}{5(1-t)} + \frac{1}{5(1-t)}$ for $t < 1$
C) $Mx(t) = \frac{2}{5(1-t)} + \frac{4}{5(1-t)}$ for $t < 1$
D) $Mx(t) = \frac{4}{5(1-t)} + \frac{2}{5(2-t)}$ for $t < 1$
E) $Mx(t) = \frac{4}{5(1+t)} + \frac{2}{5(1+t)}$ for $t > -1$
- ii) Hence, find the mean and variance of X using the moment generating function in part (i). [3]
- A) Mean = 0.8, Variance = 0.89
B) Mean = 0.9, Variance = 0.04
C) Mean = 0.8, Variance = 0.56
D) Mean = 0.8, Variance = 0.80
E) Mean = 0.9, Variance = 0.89

Q. 11) Consider the following discrete distribution with an unknown parameter q for the distribution of the number of cars with 0, 1, 2, or more than 2 accidents per year in a town that has n number of cars.

No. of accidents	0	1	2	More than 2
Probability	$4q$	$2q$	q	$1 - 7q$

Suppose X_0 is the number of cars with no accidents, X_1 is the number of cars with one accident and X_2 is the number of cars with two accidents per year. The random variable $X = X_0 + X_1 + X_2$, is defined as the number of cars with at most two accidents.

- i) Derive an expression for the maximum likelihood estimator $\hat{q}$ of parameter q in terms of X and n . [3]

- A) $\frac{x}{n}$
- B) $\frac{x}{7n}$
- C) $\frac{n}{7x}$
- D) $\frac{1-x}{7}$
- E) $\frac{x_0+x_1}{7n}$

ii) Calculate an expression for CRLB by calculating the second derivative of the log likelihood in part (i) **[4]**

- A) $\frac{1}{nq + \frac{n}{(1-q)^2}}$
- B) $\frac{1}{\frac{49n}{q} + \frac{7n}{1-7q}}$
- C) $\frac{1}{nq(1-q)}$
- D) $\frac{1}{\frac{n}{q^2} + \frac{n}{(1-q)^2}}$
- E) $\frac{q(1-7q)}{7n}$

Q. 12) The number of claims per month, X , arising on a certain portfolio of insurance policies is to be modelled using a modified geometric distribution with probability density given by:

$$p(y|\alpha) = \frac{\alpha^{y-1}}{(1+\alpha)^y} \quad y = 1, 2, 3, \dots$$

i) Express the above distribution such that X belongs to an exponential family of distributions. **[3]**

- A) $p(y|\alpha) = \exp(y \log(1 + \alpha) - \log(\alpha))$
- B) $p(y|\alpha) = \exp(y \log\left(\frac{\alpha}{1+\alpha}\right) - \log(\alpha))$
- C) $p(y|\alpha) = \exp((y-1)\log(\alpha) - y \log(1-\alpha))$
- D) $p(y|\alpha) = \exp(\alpha y - (1+\alpha)^y)$
- E) $p(y|\alpha) = \exp(y \log(\alpha) - 1)$

ii) Suggest the value of the natural parameter θ and $b(\theta)$. **[2]**

- A) $\theta = \log\left(\frac{\alpha}{1+\alpha}\right), b(\theta) = \ln\left(\frac{e^\theta}{1-e^\theta}\right)$
- B) $\theta = \log(\alpha + 1), b(\theta) = \ln(\alpha^2)$
- C) $\theta = \alpha, b(\theta) = \log(1 + \theta)$
- D) $\theta = \log\left(\frac{1+\alpha}{\alpha}\right), b(\theta) = \log(\alpha)$

E) $\theta = \log\left(\frac{\alpha}{1+\alpha}\right), b(\theta) = -\log(1 - \alpha)$

Q. 13) A discrete probability distribution is defined by:

$$f(y) = \frac{y^{\alpha-1} e^{-\frac{y}{\theta}}}{\theta^{\alpha} \Gamma(\alpha)}$$

i) Identify the canonical link function for this distribution. **[2]**

A) $g(\mu) = \mu^{\frac{1}{2}}$

B) $g(\mu) = \log(\mu)$

C) $g(\mu) = \mu$

D) $g(\mu) = \frac{1}{\mu}$

E) $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$

ii) Suggest a natural choice of link function if instead the response variable followed a lognormal distribution. **[2]**

A) $g(\mu) = \mu^{\frac{1}{2}}$

B) $g(\mu) = \log(\mu)$

C) $g(\mu) = \mu$

D) $g(\mu) = \frac{1}{\mu}$

E) $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$

Q. 14) A study was conducted to investigate hours spent on mobile phone by two teenage groups. Group A had teenagers who were not involved in any sports. Group B had teenagers who were involved in at least one sport. Let us define the number of hours spent on mobile phone by Group A by 'x' and Group B by 'y'. Independent random samples of 35 children from group A and 30 children from group B were selected. The hours spent on mobile screen by the selected group are given below:

$$\Sigma x = 205.45, \Sigma x^2 = 1245.27, \Sigma y = 120.81, \Sigma y^2 = 506.83$$

The observations from group A are assumed to be normally distributed with mean μ_1 and standard deviation σ_1 . Independently, the observations from Group B are assumed to be normally distributed with mean μ_2 and standard deviation σ_2 .

Suppose, $\sigma_1 = 1.5$ hours and $\sigma_2 = 1$ hour. Construct a 95% confidence interval for the difference between the mean length of stay for group A and group B, i.e., $\mu_1 - \mu_2$. **[4]**

A) (-2.45, -1.23)

B) (1.33, 2.35)

C) (1.23, 2.45)

D) (1.35, 2.33)

E) (1.65, 2.03)

- Q. 15)** The table below presents the percentage change in trade tariffs across eight countries, along with the corresponding change in trade deficit (in billion dollars) observed for each country in the next year. The results are given below:

Tariff Change (%) (x)	-3	-1	0.5	1.5	2	3	5	6
Trade Deficit Change \$b (y)	2.5	-1.5	-1	0.8	1.2	1.8	2.7	3

$$\Sigma x = 14, \Sigma x^2 = 86.5, \Sigma y = 9.5, \Sigma y^2 = 31.11, \Sigma xy = 34$$

- i) Calculate the correlation coefficient between Tariff percentage change and trade deficit change. [2]
- A) 0.556
B) 0.496
C) -0.73
D) 0.151
E) 0.912
- ii) You can assume that the joint distribution of the two random variables is a bivariate normal distribution. Perform a statistical test for the null hypothesis such that the true correlation coefficient between the tariff percentage change and trade deficit change is equal to 0.5, against the alternative that the correlation coefficient is smaller than 0.5. Calculate the approximate p-value using a 5% significance level. [4]
- A) 0.495
B) 0.31
C) 0.062
D) 0.21
E) 0.045
- Q.16)** A bakery owner wants to understand how the number of social media posts about their shop influences daily sales. Over a month (assume 30 days), they recorded the number of posts each day and the corresponding sales revenue. Let us define the number of social media posts by 'x' and the daily sales by 'y'. The owner wants to use linear regression to predict sales based on social media activity.

$$\Sigma x = 89, \Sigma x^2 = 478, \Sigma y = 348.2, \Sigma y^2 = 4681.13, \Sigma xy = 1323.8$$

A linear model of the form $y = \alpha + \beta x + e$ is fitted to the data, where the error terms (e) independently follow a $N(0, \sigma^2)$ distribution with σ^2 being an unknown parameter.

- i) Determine the fitted line of the regression model. [3]
- A) $\hat{y} = 3.022 + 1.359x$
B) $\hat{y} = 11.6 + 0.5x$
C) $\hat{y} = 2.345 - 1.7x$
D) $\hat{y} = 7.578 + 1.359x$
E) $\hat{x} = 4.5 + 0.5y$
- ii) Calculate a 95% confidence interval for the underlying slope coefficient for the linear model in part (i). [4]
- A) (0.945, 1.773)

- B) (1.275, 1.443)
- C) (0.963, 1.755)
- D) (-4.706, 7.424)
- E) (0.833, 1.224)

A partially completed ANOVA table for this regression analysis is given below.

Source of Variation	Degrees of Freedom	Sums of Squares	Mean Squares
Regression	A	395.29	C
Residual	B	246.15	D
Total	29	641.44	

- iii) Determine the missing values A, B, C and D in the table. [2]
- A) A = 2, B = 27, C = 197.65, D = 9.12
 - B) A = 1, B = 28, C = 197.65, D = 123.08
 - C) A = 1, B = 28, C = 246.15, D = 14.12
 - D) A = 1, B = 28, C = 395.29, D = 8.79
 - E) A = 1, B = 29, C = 395.29, D = 8.79
- iv) Determine the value of R^2 for the regression model fitted here, using the above table. [2]
- A) 28.05%
 - B) 61.63%
 - C) 38.37%
 - D) 85.02%
 - E) 90.00%
- v) Perform an F test to test the null hypothesis such that there is no linear relationship between x and y, based on the above table. Calculate the test statistic and select the right option that suggests whether to accept or reject the null hypothesis. [4]
- A) 22.49, Reject the null hypothesis
 - B) 0.62, Accept the null hypothesis
 - C) 22.12, Accept the null hypothesis
 - D) 44.97, Reject the null hypothesis
 - E) 14.28, Accept the null hypothesis

- Q. 17) A startup delivery company is trying to understand what influences the number of daily delivery delays in their city operations.

Traffic Congestion (X_1): A numerical value representing traffic severity in the city. This can take one of 3 values.

Temperature (X_2): Daily temperature in degrees Celsius. where the three categories are 10-20 degree Celsius, 20-30 degree Celsius and 30+ degree Celsius.

It wishes to model delivery delays using this data. The delivery company is investigating models which consider these variables and has the following table of values:

Model	Choice of Predictor	Scaled Deviance
A	1	220
B	Temperature	180
C	Temperature + Traffic	130
D	Temperature * Traffic	125

Explain, by analysing the scaled deviances at 5% level of significance:

i) Compare model A and B and select from the following that applies. [2]

- A) Test Statistic: 40, Critical Value: 5.991, Model A is better than model B
- B) Test Statistic: 40, Critical Value: 3.84, Model A is better than model B
- C) Test Statistic: 60, Critical Value: 5.991, Model B is better than model A
- D) Test Statistic: 40, Critical Value: 5.991, Model B is better than model A
- E) Test Statistic: 60, Critical Value: 4.61, Model A is better than model B

ii) Compare model B and C and select from the following that applies. [2]

- A) Test Statistic: 50, Critical Value: 5.991, Model C is better than model B
- B) Test Statistic: 40, Critical Value: 3.84, Model B is better than model C
- C) Test Statistic: 60, Critical Value: 5.991, Model C is better than model B
- D) Test Statistic: 40, Critical Value: 5.991, Model C is better than model B
- E) Test Statistic: 50, Critical Value: 4.61, Model B is better than model C

iii) Compare model C and D and select from the following that applies. [2]

- A) Test Statistic: 10, Critical Value: 5.991, Model D is better than model C
- B) Test Statistic: 10, Critical Value: 3.84, Model D is better than model C
- C) Test Statistic: 5, Critical Value: 9.488, Model C is better than model D
- D) Test Statistic: 5, Critical Value: 5.991, Model D is better than model C
- E) Test Statistic: 15, Critical Value: 4.61, Model C is better than model D

Q.18) An insurer's portfolio consists of three independent policies. Each policy can give rise to at most one claim per month, which occurs with probability θ independently from month to month. The prior distribution of θ is Beta with parameters α and β . A total of x claims is observed on this portfolio over an n -month period.

i) Derive the posterior distribution of θ . [2]

- A) $\theta|x \sim \text{Gamma}(\alpha, x + \beta)$
- B) $\theta|x \sim \text{Beta}(x + \alpha, x + \beta)$
- C) $\theta|x \sim \text{Gamma}(x + \alpha, 3n - x + \beta)$
- D) $\theta|x \sim \text{Beta}(x + \alpha, 3n - x + \beta)$
- E) $\theta|x \sim \text{Beta}(x + \alpha, n - x + \beta)$

ii) Find the expression for the Bayesian estimate for θ under squared error loss. [2]

- A) $\frac{\alpha}{x+\beta}$

B) $\frac{x+\alpha}{3n+\alpha+\beta}$

C) $\frac{x+\alpha}{n+\alpha+\beta}$

D) $\frac{x+\alpha}{\alpha+\beta}$

E) $\frac{\alpha}{\alpha+\beta}$

iii) Find the expression for Z, the credibility estimate.

[2]

A) $Z = \frac{3n}{3n+\alpha+\beta}$

B) $Z = \frac{n}{n+\alpha+\beta}$

C) $Z = \frac{3n}{\alpha+\beta}$

D) $Z = \frac{1}{3n+\alpha+\beta}$

E) $Z = \frac{3n}{n+\beta}$

The statistician decides to use a prior distribution of the form above with parameters $\alpha = 2$ and $\beta = 4$. You are given the following summary statistics from the sample data: $X = 9$ and $n = 12$.

iv) Calculate the value of the credibility factor, Z.

[2]

A) 0.8571

B) 0.667

C) 1

D) 0.286

E) 0.818

Q.19) Five years ago, a tech insurance company began offering coverage for electronic devices. The company classifies devices into three risk categories based on their type and usage: high-end laptops (Category 1), smartphones (Category 2), and other portable electronics (Category 3). The number of claims n_{ij} in the i^{th} category in the j^{th} year is assumed to have a Poisson distribution with unknown parameter θ_i . Data on the number of claims in each category over the last 5 years are set out as follows:

Category	Year 1	Year 2	Year 3	Year 4	Year 5	Σn_{ij}	Σn^2_{ij}
Category 1	35	42	48	55	59	239	11,875
Category 2	41	47	53	60	67	268	14,788
Category 3	24	29	20	35	31	139	4,111

Calculate the expected claims for Year 6 of each category under the assumptions of Empirical Bayes Credibility Theory Model 1.

- i)** Category 1: [2]
A) 49.20
B) 46.20
C) 45.44
D) 47.57
E) 43.07
- ii)** Category 2: [2]
A) 52.92
B) 53.12
C) 43.07
D) 45.48
E) 48.34
- iii)** Category 3: [2]
A) 43.07
B) 42.40
C) 31.00
D) 29.50
E) 28.50
